Junyan Zhou

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Education

East China Normal University Shanghai, China
B.S. in Computer Science and Technology *GPA: 3.82 (current)* Sep. 2024 – Jun. 2028 (Expected)

Technical Skills

Programming Languages:	C++, Python, HTML, HLSL, JavaScript, C#, Unity ShaderLab
Development Tools:	Git, LaTeX, PyTorch
Graphics and 3D Tools:	OpenGL, Unity, Unreal Engine, Blender
Front-End Development:	React, npm, JSX, HTML5, jQuery

Profile

Computer Science and Technology undergraduate at East China Normal University, with experience using 3D tools such as Blender and Unity and knowledge of deep learning and computer graphics. More personal projects are available on my GitHub: https://github.com/ECNUjohnny.

Projects

Personal Game Project *Unity, Blender, C#*
A 3D game developed in Unity and inspired by *Red Dead Redemption 2*. The project has a relatively complete system architecture and a range of gameplay elements.
GitHub: https://github.com/ECNUjohnny/Game-Project

Text-to-Fluid: Semantic Authoring of Fluid Simulation via Large Language Models *Unity, Unity Shader Language, C++*
This project uses an LLM to analyze the emotions conveyed by user input and express them visually by adjusting properties (such as color) of an SPH fluid rendered in Unity. I implemented a cartoon-style SPH fluid renderer and integrated an LLM to dynamically map natural-language input to rendering functions and parameters.

Text-Driven Stylized Terrain Generation *Python, PyTorch, CDSE API*
A generative model based on a variational autoencoder (VAE) and U-Net for generating stylized height maps and corresponding color textures. Integrated the Copernicus Data Space Ecosystem (CDSE) API to automate the acquisition and processing of global terrain data.
GitHub: https://github.com/ECNUjohnny/llm-style-terrain-joint

Healthcare Facility Data Visualization *React, JavaScript, JSX*
A course project for data visualization. Built a web application using React to visualize the distribution of healthcare facilities, with features including heatmaps, histograms of administrative-district statistics, and shortest-path search.
GitHub: https://github.com/ECNUjohnny/hospital-dataviz

TinyRenderer *C++*
Forked the TinyRenderer repository and implemented my own version by following the author's official tutorials. The project includes low-level C++ implementations of a rasterizer, a linear algebra library, and other fundamental components.
GitHub: https://github.com/ECNUjohnny/tinyrenderer/tree/My-branch

Research Experience

Text-to-Fluid: Semantic Authoring of Fluid Simulation via Large Language Models
International Conference on Computer-Aided Design and Computer Graphics (CAD/Graphics)
Xubin Du, Juanjie Luo, Changbo Wang, Chenhui Li, Junyan Zhou, Nan Xiang

Text-Driven Joint Generation of Terrain Geometry and Texture in a Shared Latent Space *IEEE VR (under review)*
Junyan Zhou, Yunze Han, Haizhou Zhu, Chenhui Li

Honors and Awards

Lanqiao Cup National Software and IT Professionals Competition
Shanghai Regional First Prize — April 2026
Organizer: Ministry of Industry and Information Technology of China

Lanqiao Cup National Software and IT Professionals Competition
National Finals Third Prize — June 2026
Organizer: Ministry of Industry and Information Technology of China

National College Student Mathematics Competition
Shanghai Regional First Prize — December 2025
Organizer: Chinese Mathematical Society

IELTS Academic Test — British Council
Overall Band Score: 7.5; Reading: 9.0, Listening: 8.5, Speaking: 6.5, Writing: 6.5 — July 2026